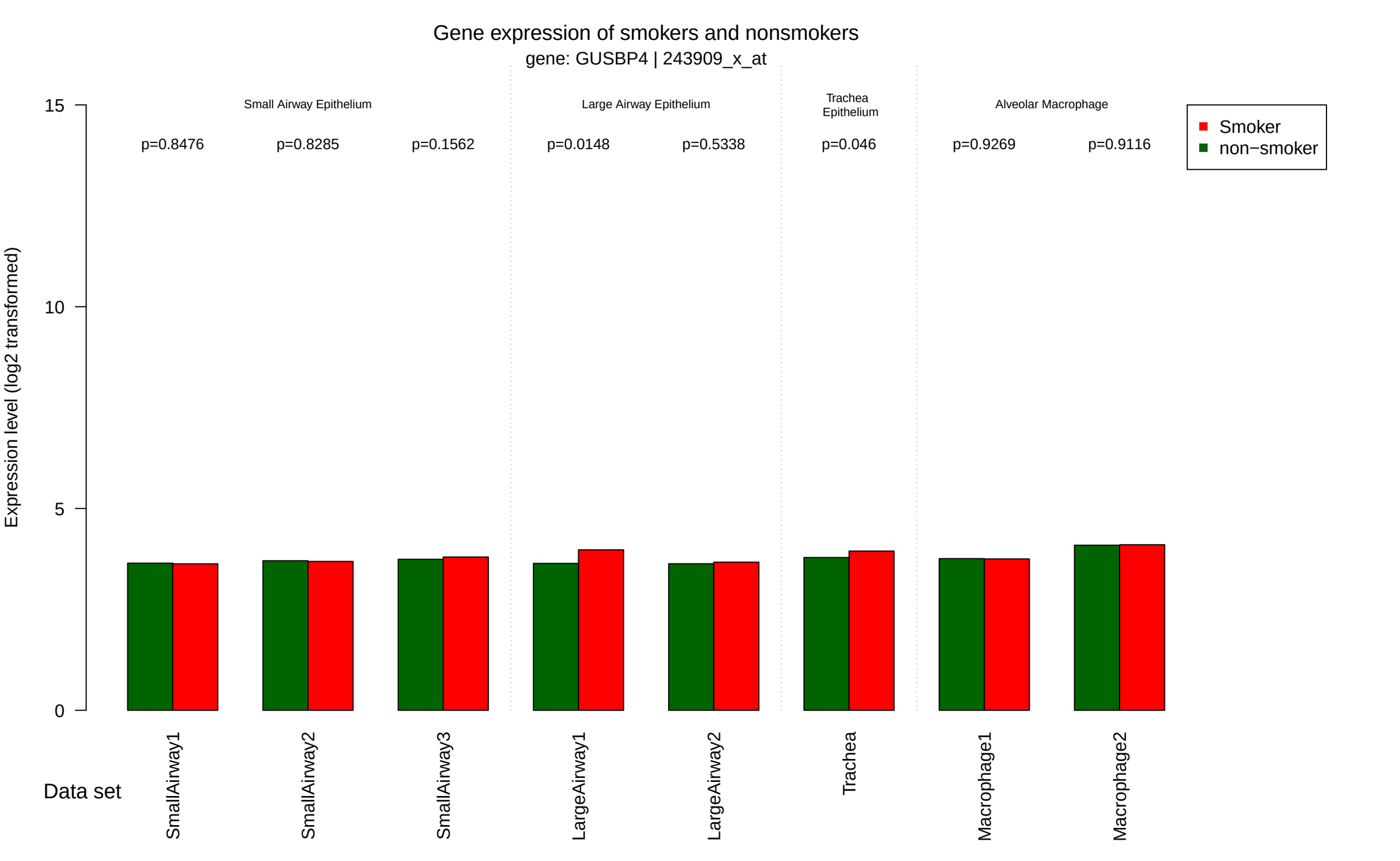
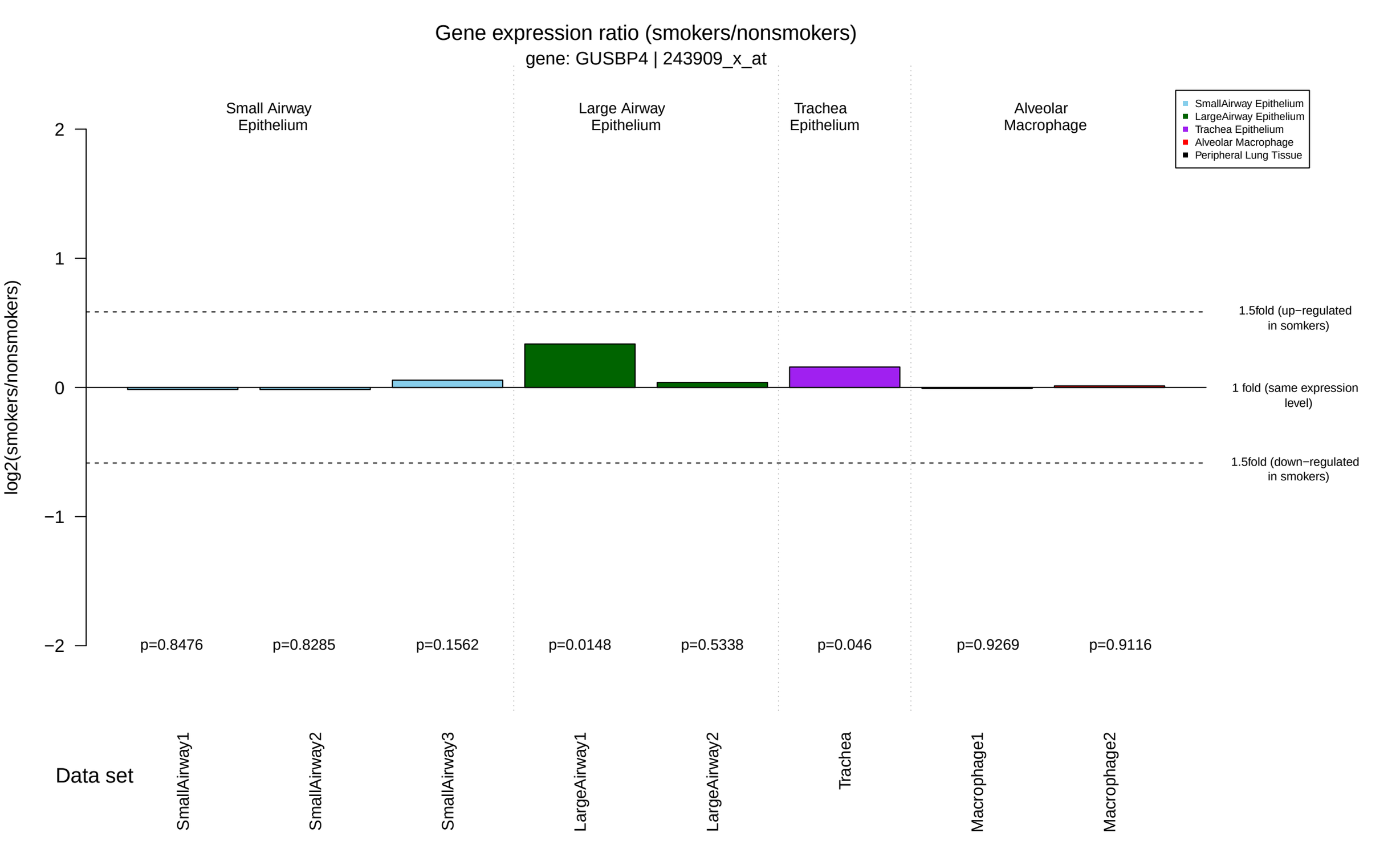
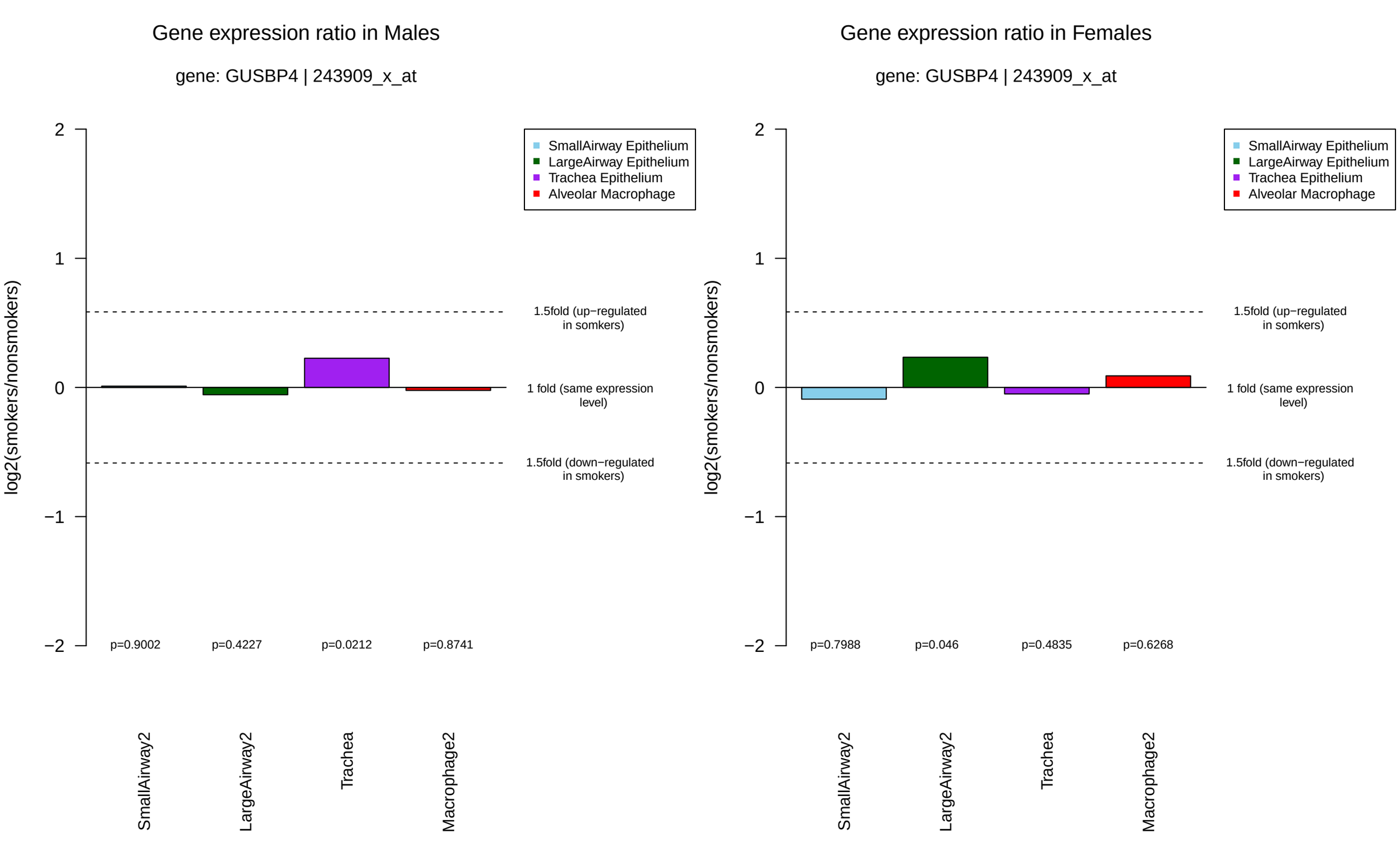
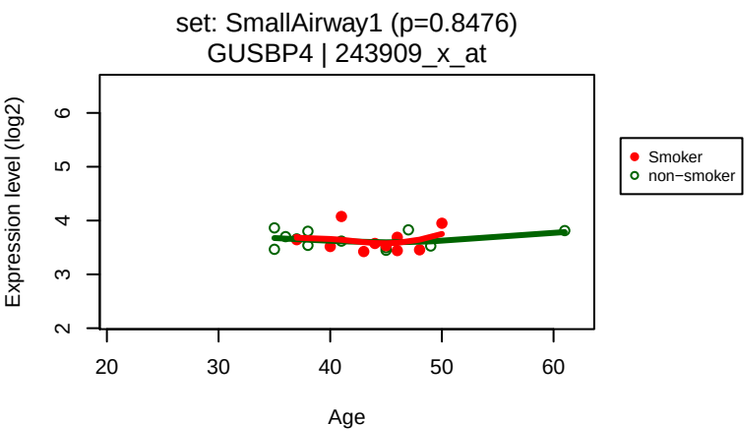
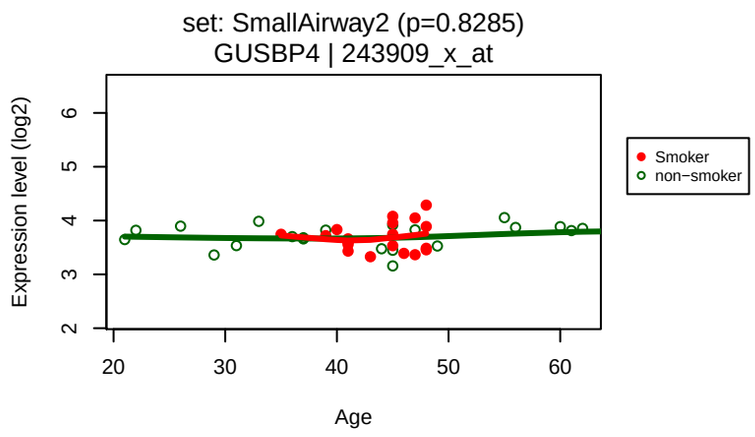
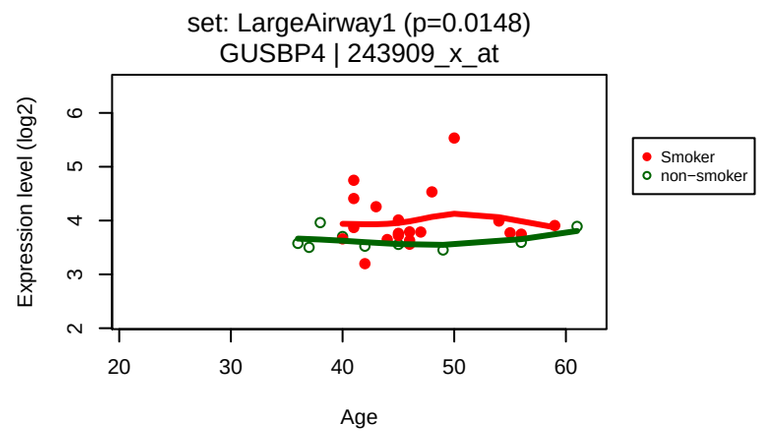

Gene expression of smokers and nonsmokers
gene: GUSBP4 | 243909_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8476
p=0.8285
p=0.1562
p=0.0148
p=0.5338
p=0.046
p=0.9269
p=0.9116
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

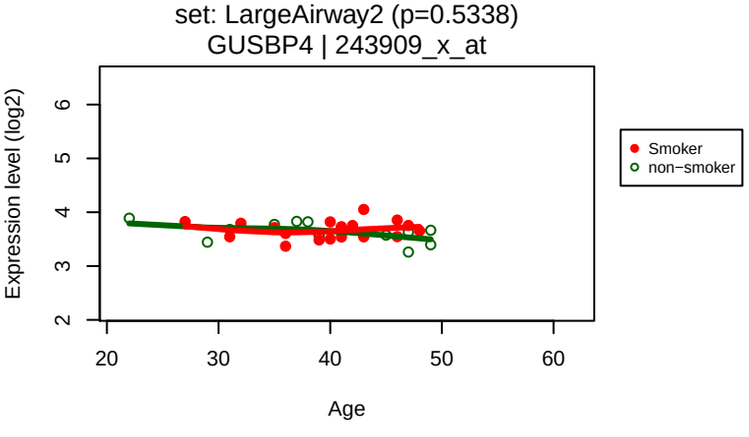

Gene expression ratio (smokers/nonsmokers)
gene: GUSBP4 | 243909_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8476
p=0.8285
p=0.1562
p=0.0148
p=0.5338
p=0.046
p=0.9269
p=0.9116
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

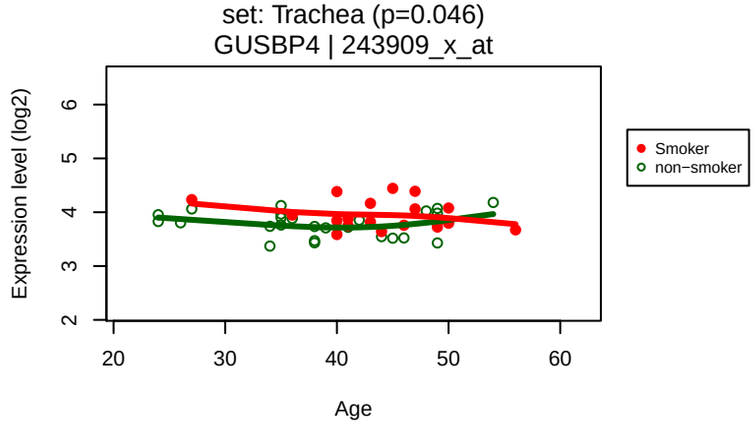

Gene expression ratio in Males
gene: GUSBP4 | 243909_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9002
p=0.4227
p=0.0212
p=0.8741
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GUSBP4 | 243909_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7988
p=0.046
p=0.4835
p=0.6268
SmallAirway2
LargeAirway2
Trachea
Macrophage2

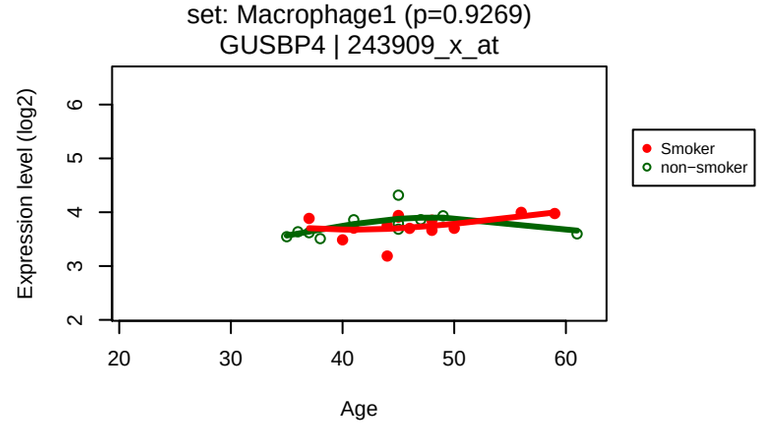

set: SmallAirway1 (p=0.8476)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker

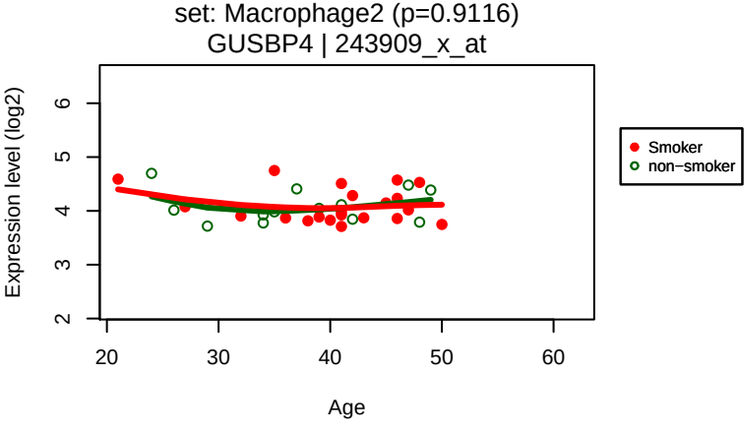

set: SmallAirway2 (p=0.8285)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker

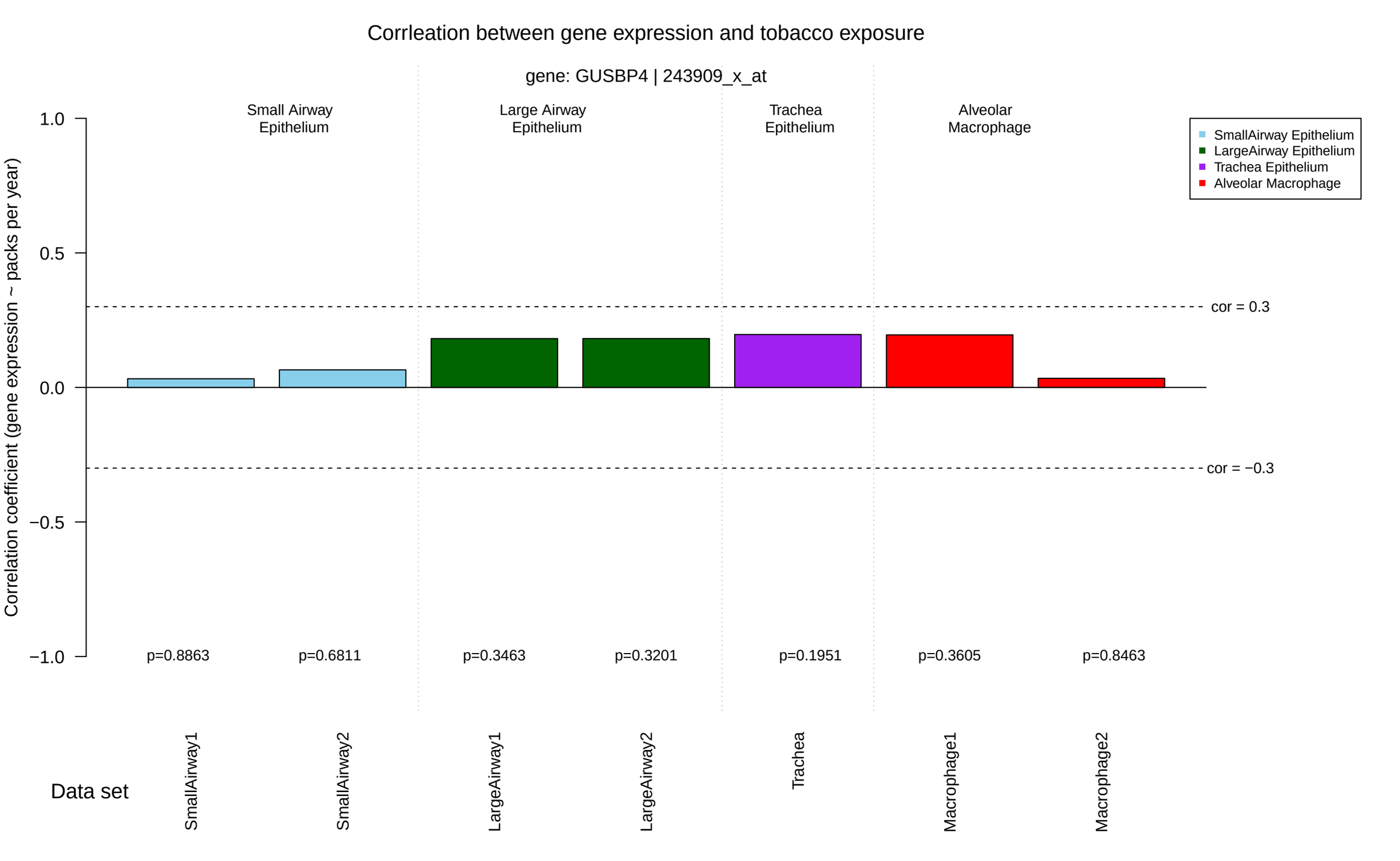

set: LargeAirway1 (p=0.0148)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5338)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.046)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9269)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9116)
GUSBP4 | 243909_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GUSBP4 | 243909_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8863
p=0.6811
p=0.3463
p=0.3201
p=0.1951
p=0.3605
p=0.8463
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2